

AI Misuse, Harmful Incidents, and Mitigation Approaches

A Summary Report

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Abstract

This document summarizes key harmful uses of AI identified through an analysis of the AI Incident Database, correlates them with systemic risks, and outlines concrete mitigation techniques derived from current AI safety research. The goal is to provide a clear, actionable framework for reducing the frequency and severity of harmful AI incidents.

1. Identified Harmful Use Cases

Based on an analysis of recent incident data, three primary themes of harmful AI use have been identified:

1.1. Deepfakes & Synthetic Media Impersonation

- **Description:** The use of AI to generate convincing but fake videos, voices, and images to impersonate real individuals, including public figures and private citizens.
- **Example Incidents:** AI voice cloning for financial scams, non-consensual deepfake content, and fraudulent celebrity endorsements.

1.2. AI-Enabled Fraud & Phishing

- **Description:** The use of Large Language Models and other generative AI to create deceptive content at scale for the purpose of stealing credentials, committing fraud, or manipulating individuals.
- **Example Incidents:** Highly personalized phishing email campaigns, automated generation of fraudulent websites, and fake investment scam promotions.

1.3. Unsafe Exposure & Algorithmic Harm

- **Description:** Incidents where AI systems, often operating as intended, produce unintended negative consequences, leading to harm or the violation of rights.
- **Example Incidents:** Recommendation algorithms promoting harmful or dangerous content to children, biased facial recognition systems leading to

wrongful arrests, and flawed algorithmic risk-scoring systems in justice and social welfare.

2. Proposed Mitigation Strategies

To address these identified harms, a multi-layered approach is required, combining technical solutions from AI labs with robust governance and policy.

2.1. Embed Forensic Detection and Traceability

- **Goal:** To counter impersonation and synthetic media misuse.
- **Actions:**
 - Implement real-time deepfake detection modules in platforms.
 - Mandate the adoption of **C2PA (Coalition for Content Provenance and Authenticity)** standards for cryptographic watermarking and tamper-evident metadata in all AI-generated content.
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- **Reference:** Mökander, J., & Schuett, J. (2025). *AI and misinformation*. Springer.

2.2. Strengthen Access Control, Monitoring & Human Oversight

- **Goal:** To reduce systemic harm from algorithmic misuse and prevent anonymous abuse of powerful AI systems.
- **Actions:**
 - Apply **Zero-Trust** security models to AI systems, enforcing continuous monitoring and segmented privileges.
 - Require a **human-in-the-loop** for review and final approval of high-risk AI outputs (e.g., in content moderation, medical diagnostics, or judicial sentencing).
 - Implement stricter **"Know Your Customer" (KYC)** protocols for access to high-risk APIs, such as those for voice cloning.
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- **Reference:** Vanta. (n.d.). *AI security best practices*.

2.3. Implement Robust Governance & Capability Restriction

- **Goal:** To reduce the overall frequency and severity of incidents by creating a safer development and deployment ecosystem.
- **Actions:**
 - Establish clear regulatory frameworks that require independent **auditing and red-teaming** of high-risk AI models before public deployment.
 - License or otherwise restrict access to the most powerful, frontier AI capabilities to prevent widespread proliferation.

- Invest in large-scale **user and enterprise awareness programs** to educate the public on identifying AI-generated content and scams.
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 - **Reference:** Stanford ESR. (n.d.). *A Framework for Risk Mitigation*.
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3. References

Bhatia, K., & Deshpande, A. (2024). *AI governance and explainable AI*. Springer Link. <https://link.springer.com/article/10.1007/s00146-024-02130-8>

Mökander, J., & Schuett, J. (2025). *AI and misinformation: A systematic review*. Springer Link. <https://link.springer.com/article/10.1007/s00146-025-02620-3>

Stanford University Ethics, Society, and Technology Hub. (n.d.). *Towards a Framework for Risk Mitigation....* Retrieved from <https://esr.stanford.edu/document/towards-a-framework-for-risk-mitigation-of-potential-misuse-of-artificial-intelligence-in-biomedical-research/>

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